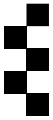
Business Intelligence and Machine Learning



University of Essex

Business Intelligence Systems

- Business intelligence (BI) systems are information systems that assist managers and other professionals
  - in the analysis of current and past activities

  and
  - in the prediction of future events

Categories of BI Systems

- **Reporting systems**
  - sort, filter, group, and make elementary calculations on operational data.
  - creates meaningful information from different data sources.
- **Data mining applications**
  - perform sophisticated analyses on data, analyses that usually involve complex statistical and mathematical processing.

Data Analytics in Business

- Data analytics is used in business to help organizations make better business decisions.
- Whether it's market research, product research, customer reviews, sentiment analysis, or any other issue for which data exists.
- Analyzing data will provide insights that organizations need in order to make the right choices.

Types of Analytics

Descriptive analytics	Predictive analytics	Prescriptive analytics
• Describes the **past status of the domain of interest** using a variety of tools through techniques such as reporting, data visualization, dashboards, and scorecards	• Applies statistical and computational methods and models to data regarding past and current events to predict **what might happen in the future**	• Uses **results of predictive analytics** along with optimization and simulation tools to **recommend actions** that will lead to a desired outcome

Problem with Data

- Problematic data are termed **dirty data**.
  - Examples are a value of “K” for customer sex and a value of “213” for customer age.
- **Inconsistent data** is particularly common for data that have been gathered over time.
  - When an area code changes, **for example**, the phone number for a given customer before the change will differ from the customer’s phone number after the change.
- Purchased data often contain **missing values**.

Machine Learning

- Machine Learning is a subset of artificial intelligence.

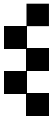
- It focuses mainly on the designing of systems which are capable to learn from the given data/information and make predictions (or do classification).

- Machine learning can be supervised or unsupervised.

Classification of Data

- Classification is the process of predicting the class of given data points.
- Classes are sometimes called as targets/ labels or categories
- Various machine learning algorithms are:
  - Decision Tree
  - Artificial Neural Networks
  - Support Vector Machine
  - Gaussian Mixture Model
  - k-means

Classification in Daily-life



University of Essex

Tennis Balls vs Golf Balls





University of Essex

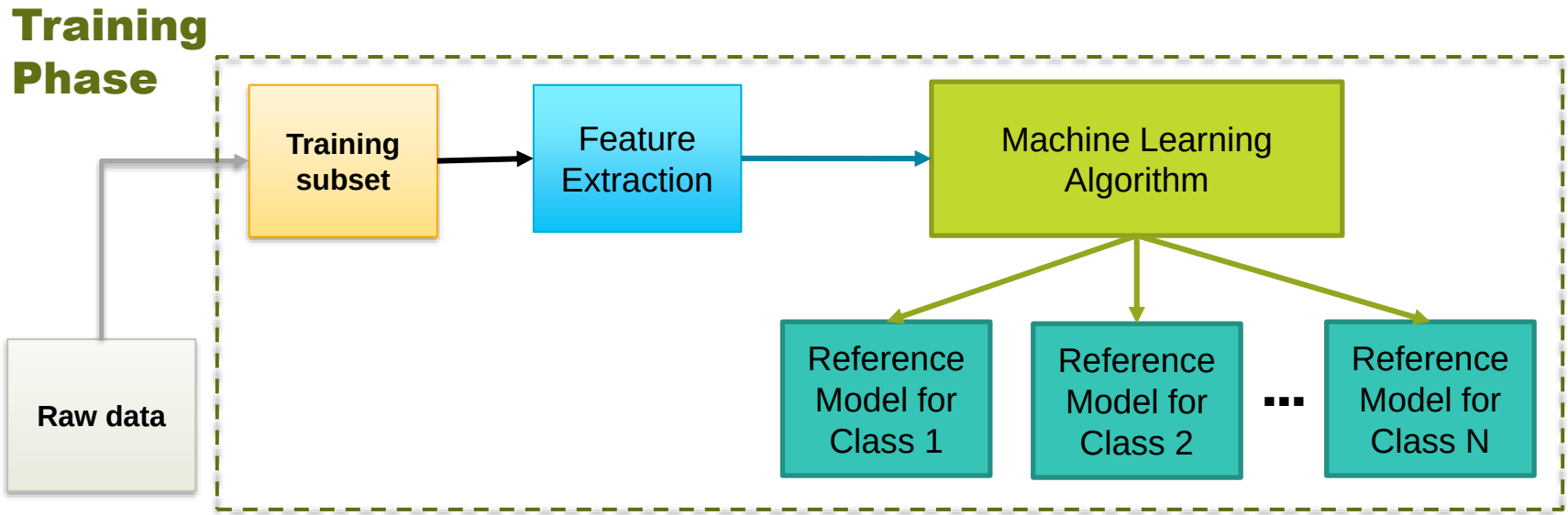
Predict Class



Predict Class

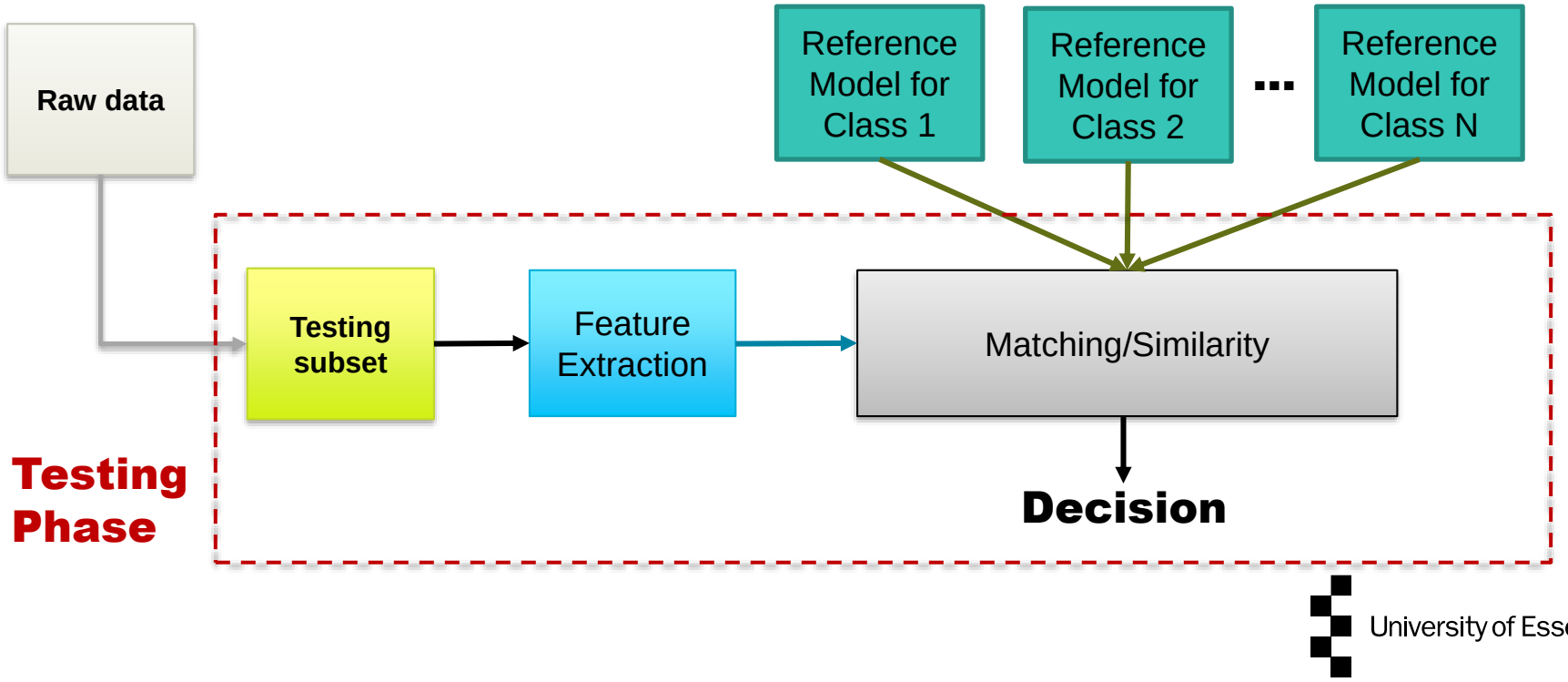


A Typical Machine Learning based Classification System



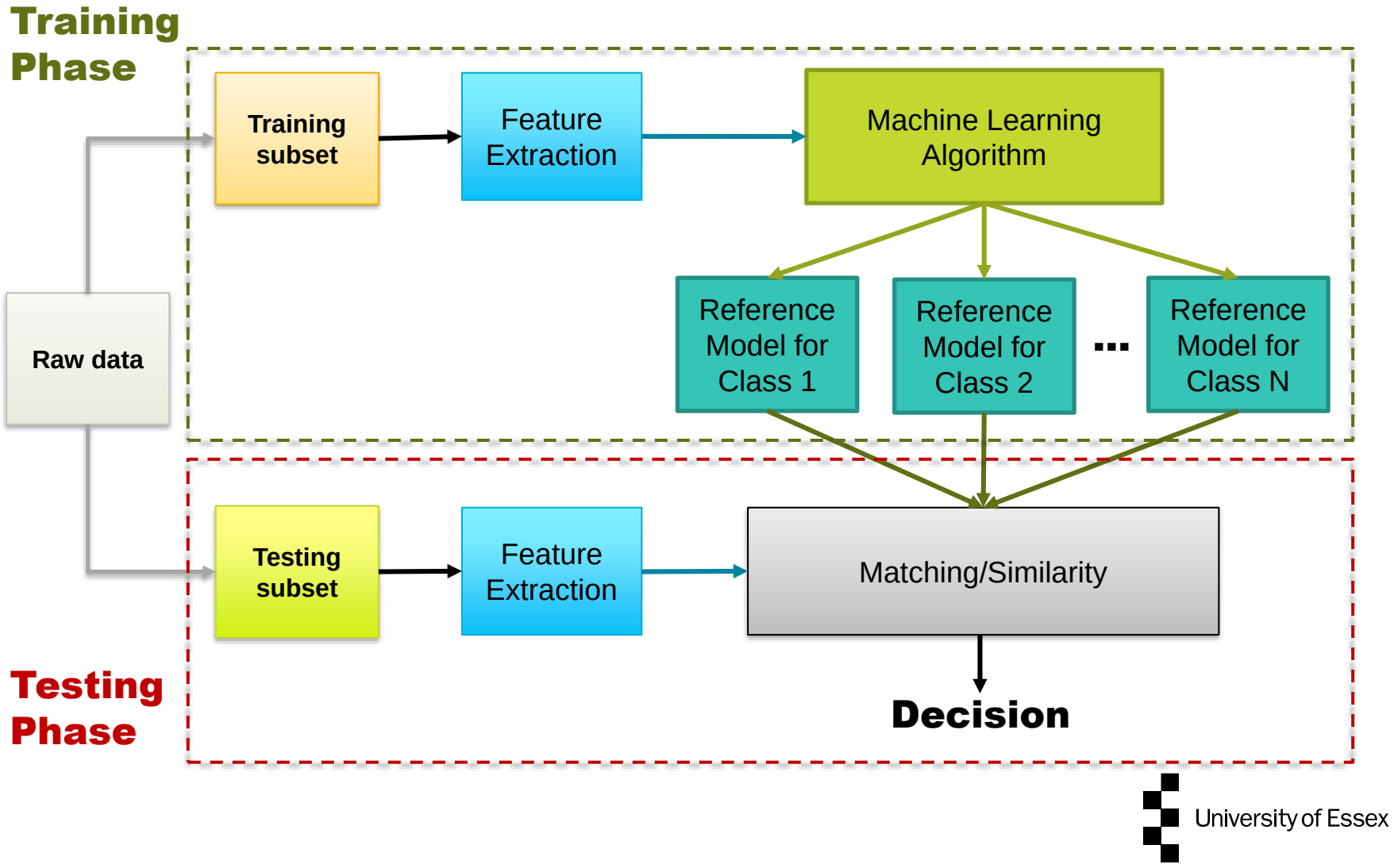
University of Essex

A Typical Machine Learning based Classification System



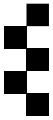
University of Essex

A Typical Machine Learning based Classification System



University of Essex

Supervised Learning



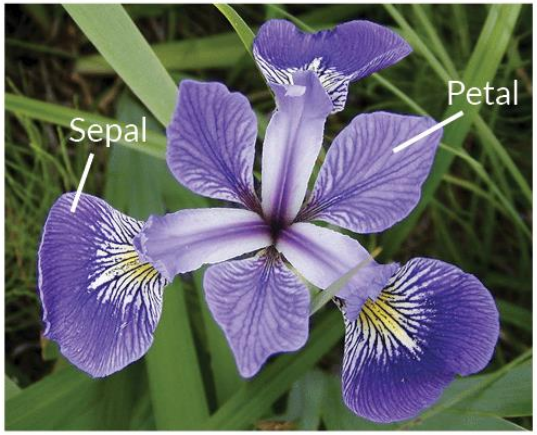
University of Essex

Supervised Learning

- Supervised learning methods assume that labels are available for all training samples.
- The training samples and labels are used to generate reference models for all classes in the dataset.
- Then, generated models are used as the basis for predicting the classification of other unlabeled data.

Iris Flower Dataset

- The Iris flower dataset or Fisher's Iris dataset is a multivariate data set introduced by the British scientist Ronald Fisher in 1936.



Iris Versicolor



Iris Setosa



Iris Virginica

- The data set consists of **50 samples** from each of three species of Iris.

University of Essex

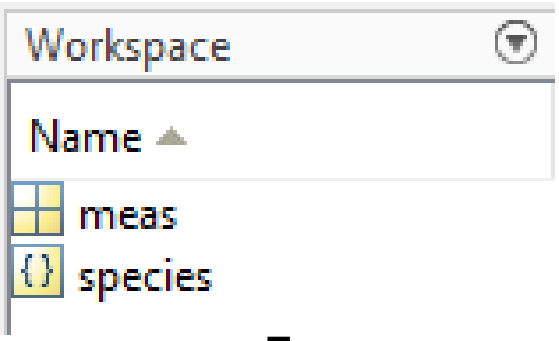
Iris Flower Dataset - How to load in MATLAB

>> load fisheriris

- Two variables will appear in the workspace
  - ***meas*** 150 x 4 (150 rows and 4 column)

    (it contains measurement of three species, 50 for each)
  - ***species*** 150x1 (150 rows and 1 column)

    (it contains label for each row of meas)



University of Essex

Variable *meas* and *species*

meas

	1	2	3	4
1	5.1000	3.5000	1.4000	0.2000
2	4.9000	3	1.4000	0.2000
3	4.7000	3.2000	1.3000	0.2000
4	4.6000	3.1000	1.5000	0.2000
5	5	3.6000	1.4000	0.2000
.....				
47	5.1000	3.8000	1.6000	0.2000
48	4.6000	3.2000	1.4000	0.2000
49	5.3000	3.7000	1.5000	0.2000
50	5	3.3000	1.4000	0.2000
51	7	3.2000	4.7000	1.4000
52	6.4000	3.2000	4.5000	1.5000
.....				
98	6.2000	2.9000	4.3000	1.3000
99	5.1000	2.5000	3	1.1000
100	5.7000	2.8000	4.1000	1.3000
101	6.3000	3.3000	6	2.5000
102	5.8000	2.7000	5.1000	1.9000
.....				
149	6.2000	3.4000	5.4000	2.3000
150	5.9000	3	5.1000	1.8000
151				

species

	1
1	setosa
2	setosa
3	setosa
4	setosa
5	setosa
.....	
47	setosa
48	setosa
49	setosa
50	setosa
51	versicolor
52	versicolor
.....	
98	versicolor
99	versicolor
100	versicolor
101	virginica
102	virginica
.....	
149	virginica
150	virginica
151	

University of Essex

Training and Testing Subsets

- Use first 30 samples of each species to generate models in training phase.

  S_train=meas(1:30, 3:4); *% first 30 samples of setosa*

  S_tr_lab=species(1:30); *% labels of first 30 samples*

- Use the remaining 20 samples of each species to evaluate the reference models in testing phase.
  - These samples will be unknow to the reference models.

  S_test=meas(31:50, 3:4); *% remaining 20 samples of setosa*

  S_ts_lab=species(31:50); *% labels of remaining 20 samples of setosa*

Generate Reference Models for Each Class

- I have implemented k-means algorithm.

```
No_clusters=3; % you can change it
opts = statset('Display','final'); % display the final iteration of k-means
```

Generate model for the first class (setosa)

```
[idxS,CS] = kmeans(S_train,No_clusters,'Options',opts);
```

Generate model for the second class (versicolor)

```
[idxV,CV] = kmeans(V_train,No_clusters,'Options',opts);
```

Components of a Generated Reference Model

Reference Model for a Class

Training Subset

	1	2
1	1.4000	0.2000
2	1.4000	0.2000
3	1.3000	0.2000
4	1.5000	0.2000
5	1.4000	0.2000
6	1.7000	0.4000
7	1.4000	0.3000
8	1.5000	0.2000
9	1.4000	0.2000
10	1.5000	0.1000
11	1.5000	0.2000
12	1.6000	0.2000
.....		
27	1.6000	0.4000
28	1.5000	0.2000
29	1.4000	0.2000
30	1.6000	0.2000

Indices of samples

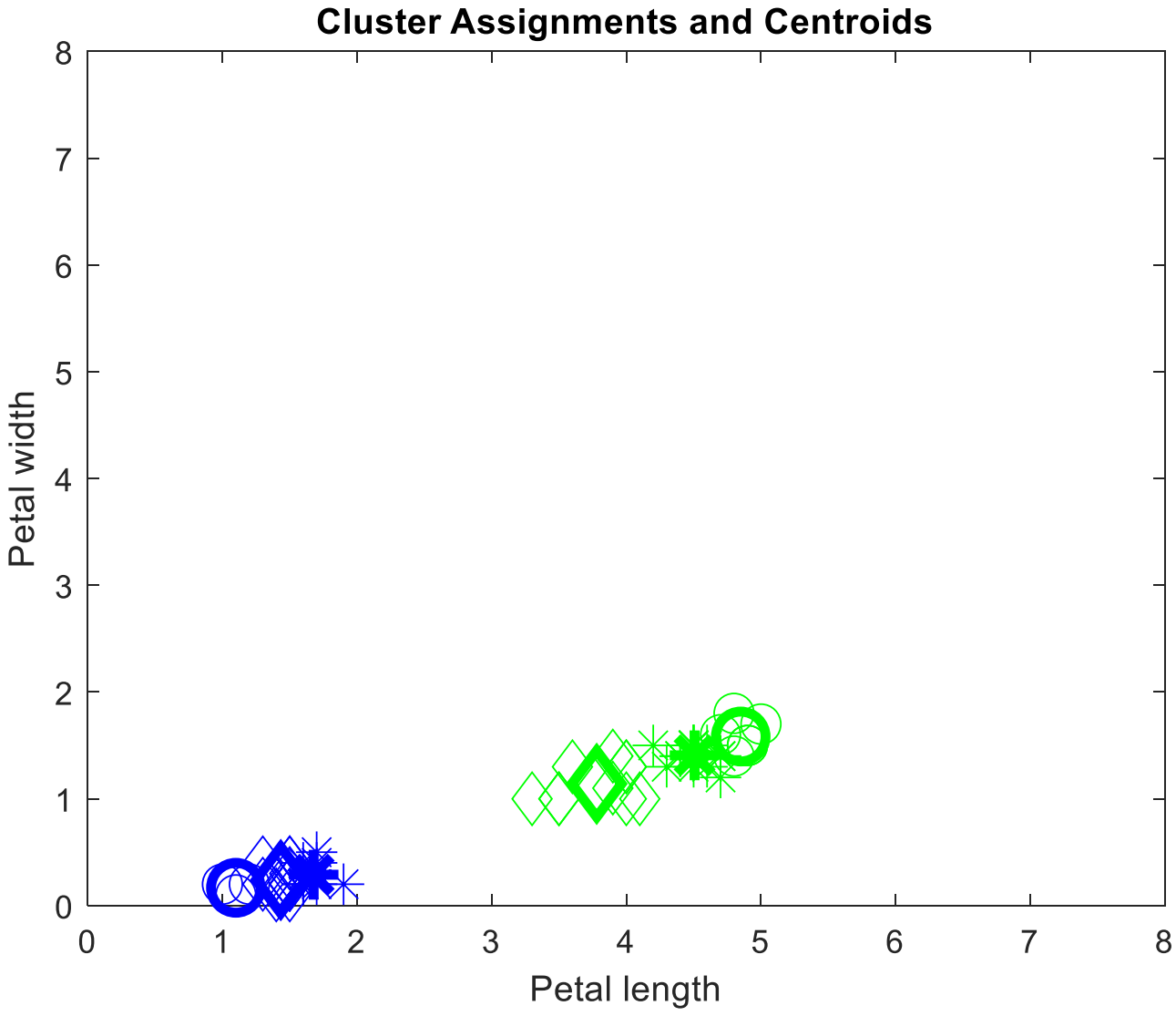
	1
1	1
2	1
3	1
4	1
5	1
6	2
7	1
8	1
9	1
10	1
11	1
12	2
.....	
27	2
28	1
29	1
30	2

Centroids (*CS*)

	1	2
1	1.4333	0.2389
2	1.6778	0.2889
3	1.1000	0.1667

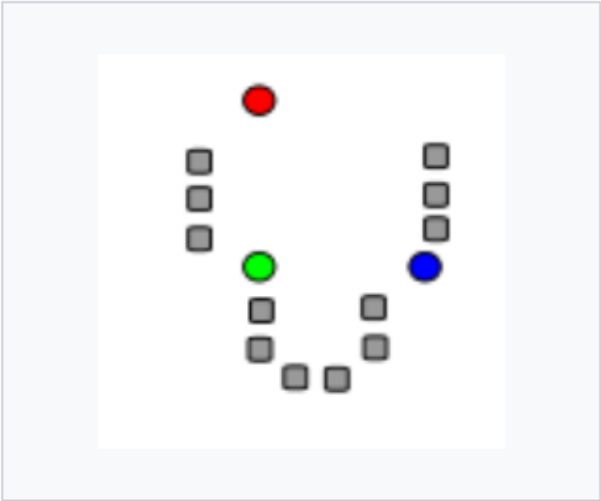
University of Essex

Visualize Clusters and Centroid

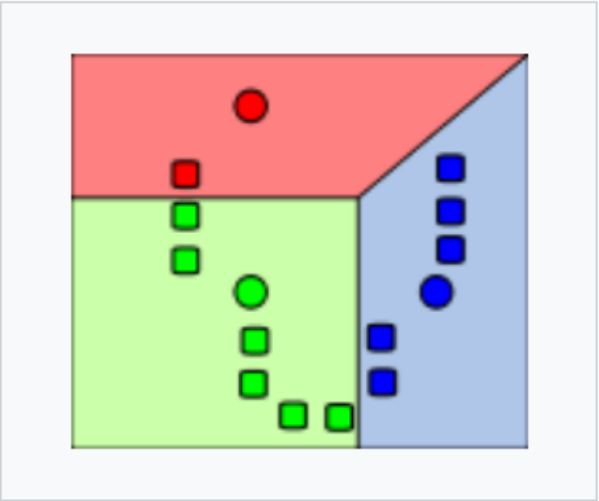


University of Essex

K-Means Algorithm (1)

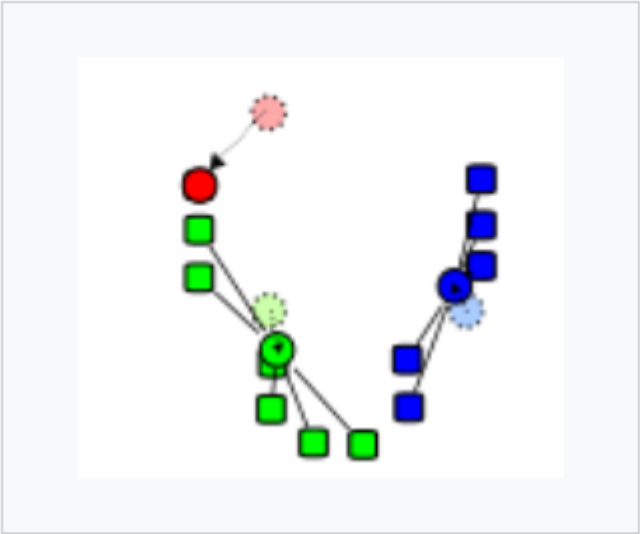


1. *k* initial "means" (in this case *k*=3) are randomly generated within the data domain (shown in color).

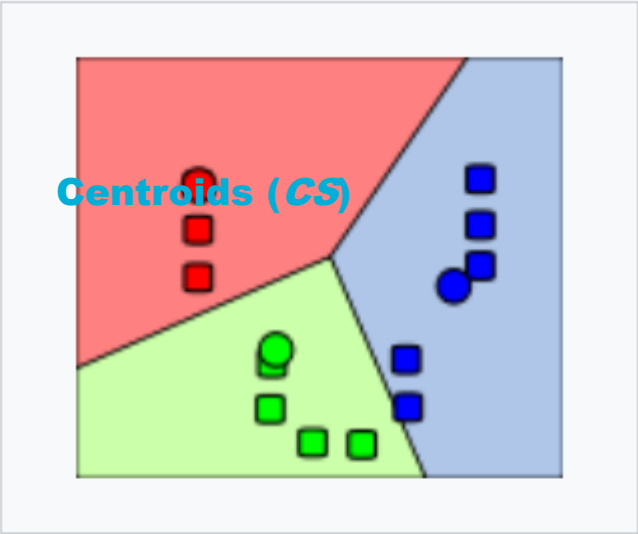


2. *k* clusters are created by associating every observation with the nearest mean.

K-Means Algorithm (2)

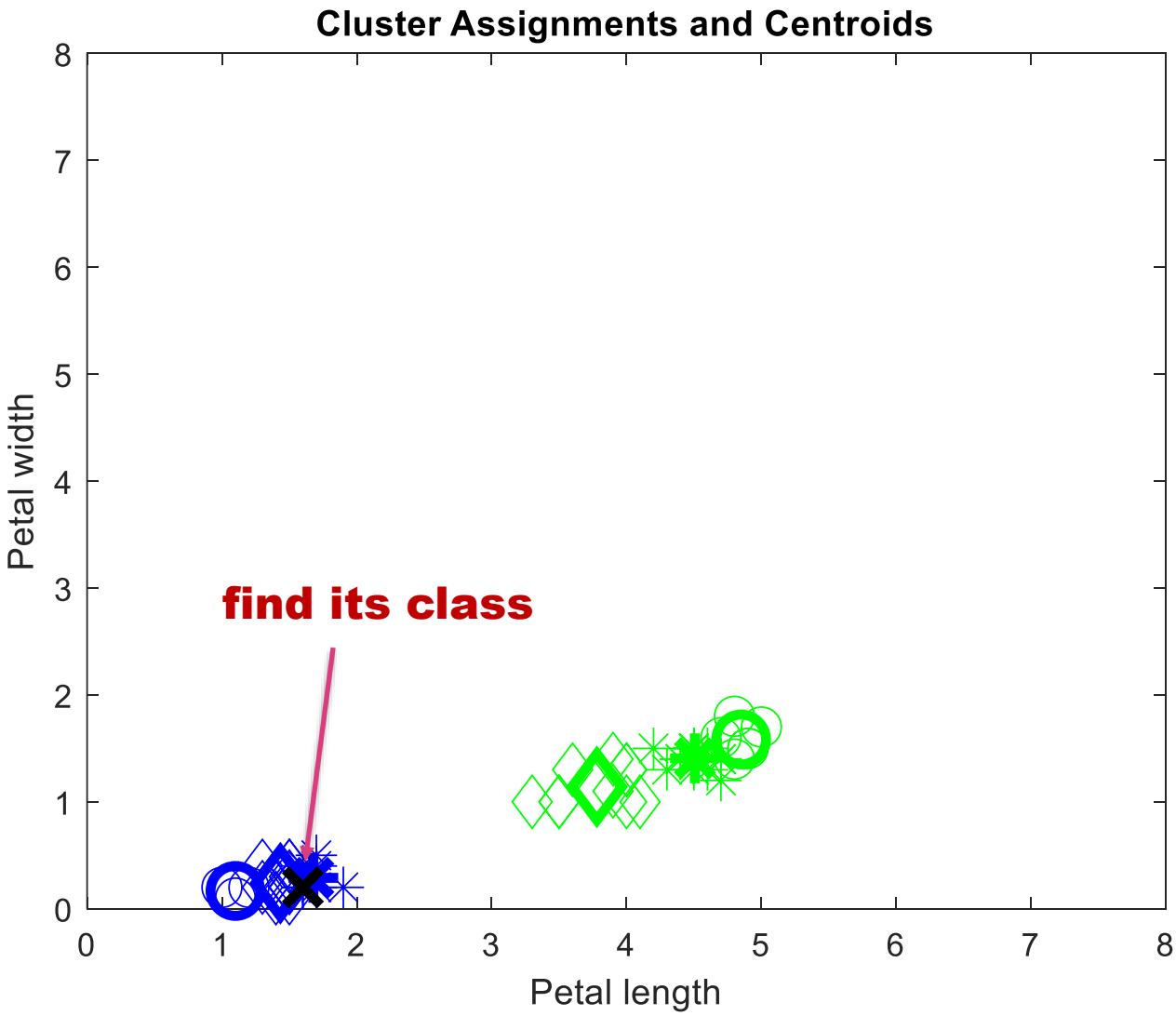


3. The centroid of each of the *k* clusters becomes the new mean.



4. Steps 2 and 3 are repeated until convergence has been reached.

Unknown Test Sample



University of Essex

Decision about a Test Sample (1)

- Take a test sample from any class
- Compute distance of the unknown sample from each centroids of each class
- You will get 3 distances (one for each centroid) for each reference model

```
DS1  = sqrt(sum((test_Sample - CS(1,:)) .^ 2));
DS2  = sqrt(sum((test_Sample - CS(2,:)) .^ 2));
DS3  = sqrt(sum((test_Sample - CS(3,:)) .^ 2));
```

Similarly, compute distances with other refence model.

Decision about Test Sample (2)

- Now compute average distance

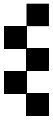
  mDS=(DS1+DS2+DS3)/3;

- Similarly, Compute average distance for the other class

  mDV=(DV1+DV2+DV3)/3;

If the minimum average distance of the unknown sample is with Setosa then predicted class will be Setosa, otherwise Versicolor.

Unsupervised Learning



University of Essex

Unsupervised Learning (1)

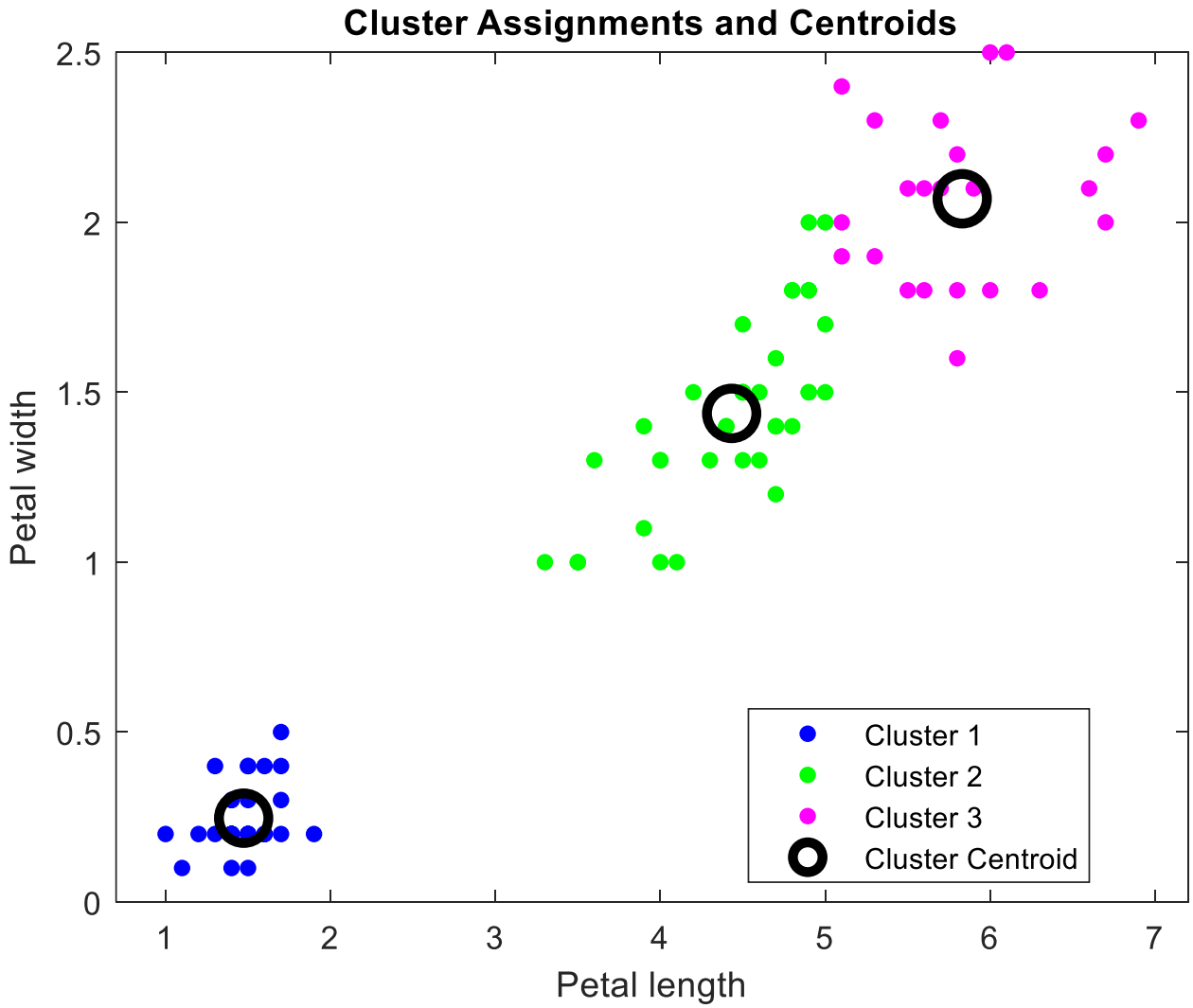
- Unsupervised learning is a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labels.

- The most common unsupervised learning method is cluster analysis, which is used for exploratory data analysis to find hidden patterns or grouping in data.

Unsupervised Learning (2)

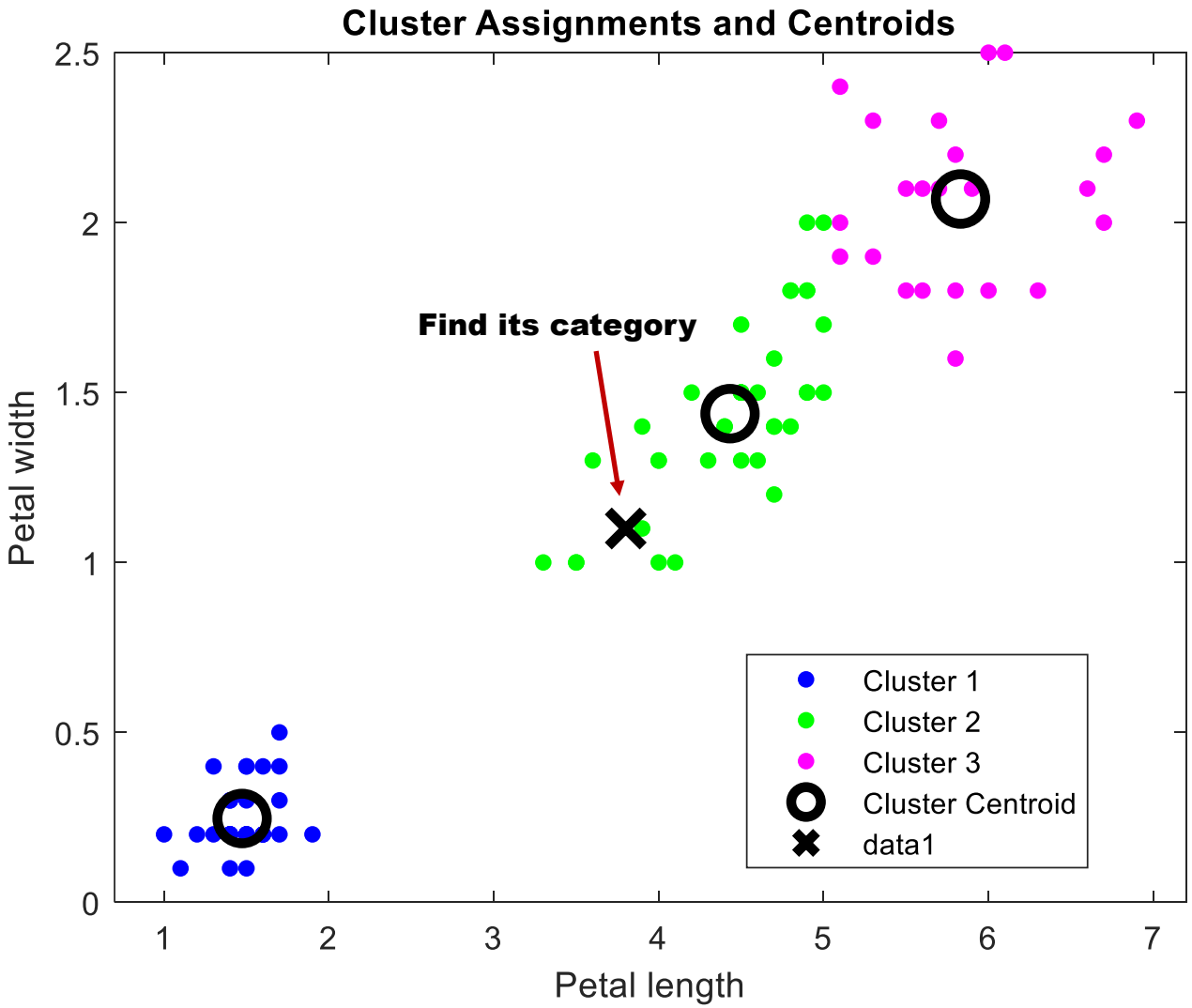
- Although, the labels of the samples will not be available, it is important to know the categories in data.
- The classification of unlabeled data Using unsupervised learning is a difficult task as compared to supervised learning.
- Also, it is difficult to know the accuracy of the generated reference models (as labels are unavailable).

Generated Reference Models using Unsupervised Learning



University of Essex

Test Sample



University of Essex

Decision

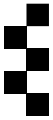
- Compare the test sample with generated reference model of each category.

```
DS1  = sqrt(sum((test_Sample - C(1,:)) .^ 2));
DS2  = sqrt(sum((test_Sample - C(2,:)) .^ 2));
DS3  = sqrt(sum((test_Sample - C(3,:)) .^ 2));
```

- Test sample will belong to a cluster which has a minimum distance with it.

```
[values, ind]=min([DS1 DS2 DS3]);
```

Evaluation of Classification System



University of Essex

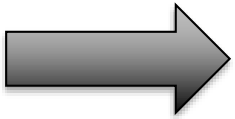
How to find TP, FN, TN and FP?

What is your positive class?

Let's say, it is ***setosa.***

TP: True Positive
FN: False Negative
TN: True Negative
FP: False Positive

Original Labels	Predicted by System	
setosa	setosa	TP
setosa	versicolor	FN
setosa	setosa	TP
versicolor	setosa	FP
versicolor	versicolor	TN



University of Essex

Some Performance Measures for Evaluation

sensitivity = true positive rate

$$= (TP / totalPositive) = \frac{TP}{TP + FN}$$

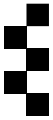
specificity = true negative rate

$$= (TP / totalNegative) = \frac{TN}{TN + FP}$$

accuracy = (trulyPredictedSamples / totalSamples)

$$= \frac{TP+TN}{TP+FN+TN+FP}$$

Feature Selection



University of Essex

Fisher's Discriminant ratio

$$FDR = \frac{(\mu_P - \mu_N)^2}{\sigma_P^2 + \sigma_N^2}$$

where

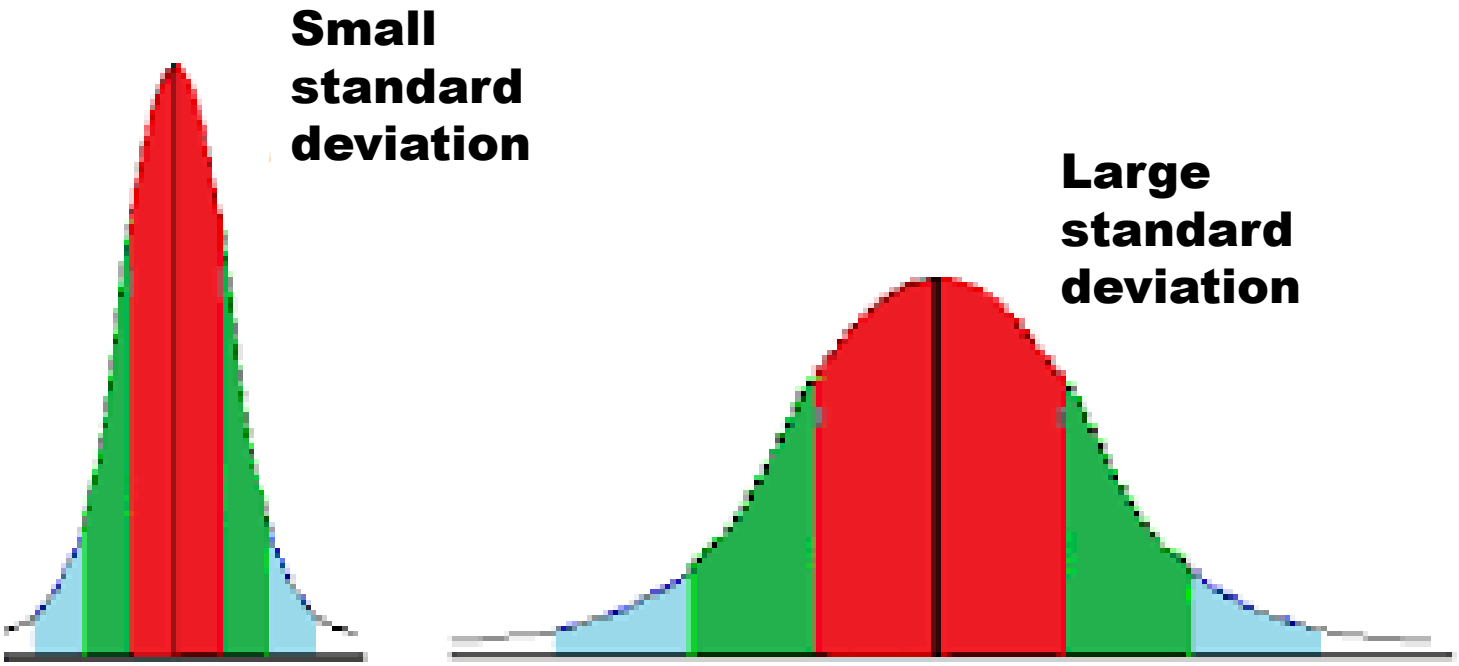
- μ_P and μ_N represent **means** of positive and negative classes for the corresponding features.
- Similarly, σ_P and σ_N represent **standard deviation** of positive and negative classes for the corresponding features.

Fisher's Discriminant ratio

- If **FDR is high**, the corresponding features are **good** to discriminate between the given classes.
- If **FDR is low**, the corresponding features **are not good** to discriminate between the given classes.

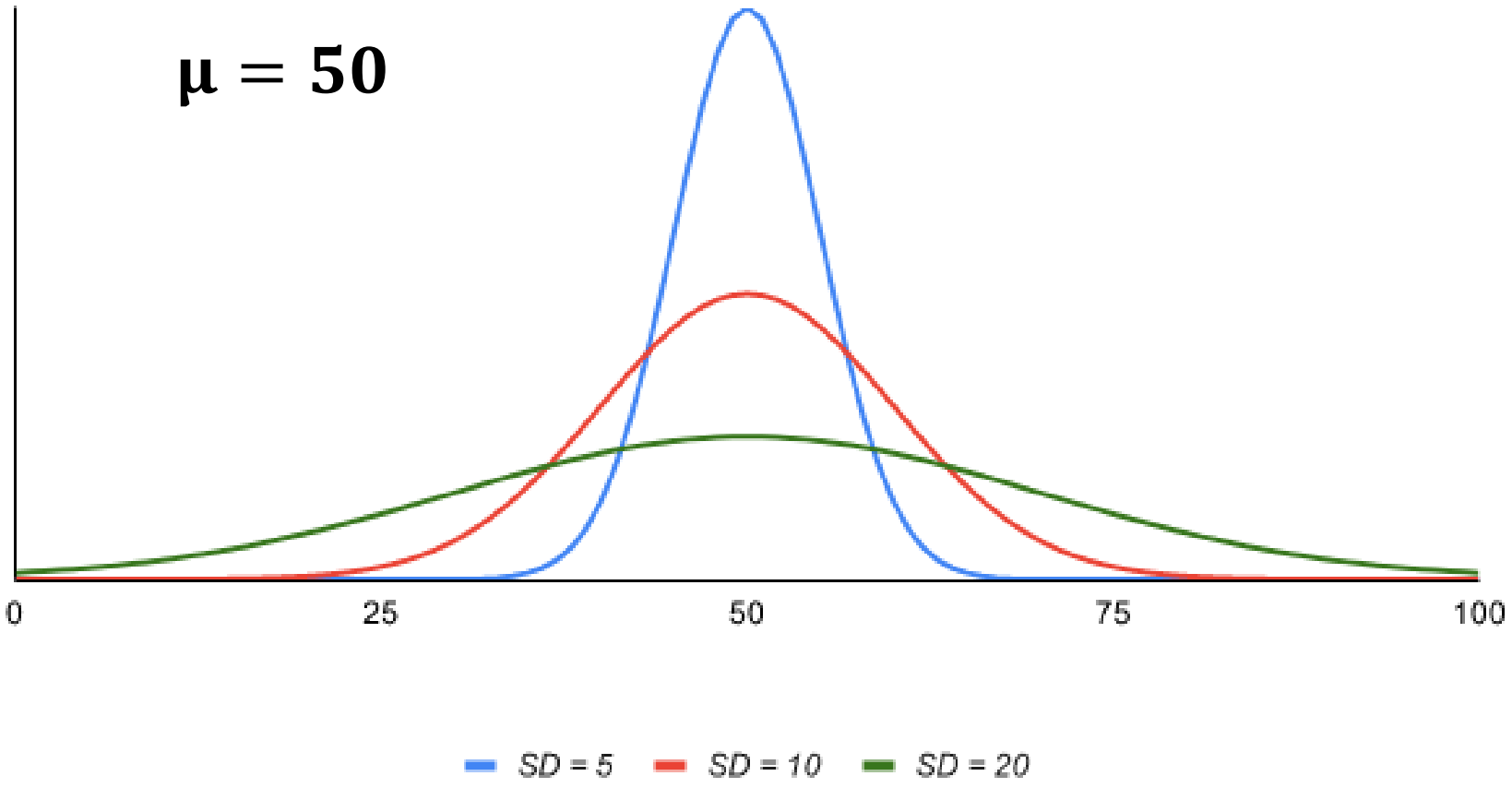
$$FDR = \frac{(\mu_P - \mu_N)^2}{\sigma_P^2 + \sigma_N^2}$$

σ to find Spread of a data (Feature) (1)



University of Essex

σ to find Spread of data (Feature) (2)



University of Essex